National University of Computer and Emerging Sciences [redacted]

Feedback Control Systems (EE3004)

Date: May 29th 2025

Course Instructor

Dr. Omer Saleem and Ms. Tamania Javaid

Final Exam

Total Time: 3 Hours

Total Marks: 100

Total Questions: 05

Semester: SP-2025

Campus: Lahore

Dept: Electrical Engg.

Student Name ________ Roll No ________ Section ________ Student Signature ________

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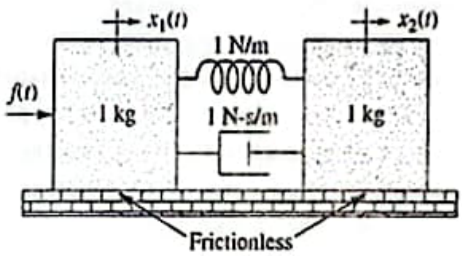
CLO # 1: Develop the mathematical model of a physical system.

Q1: **[20 marks]**

a. Develop the state space representation of the following mechanical system.



b. Formulate transfer function of the form, Y(s)/U(s), using the following state space representation. Hence, find the expression of output, $y(t)$, if the input signal, $u(t)$, is a unit-step signal.

$$\dot{x}(t) = \begin{bmatrix} -5 & 0 \\ 0 & -1 \end{bmatrix} x(t) + \begin{bmatrix} 3 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 7 & 0 \end{bmatrix} x(t) + [0]\, u(t)$$

CLO # 2: Analyze the transient and steady state response of a feedback system.

Q2: **[20 marks]**

a. Analyze the closed-loop system shown below.

i. Reduce the system's block diagram to find its overall transfer function T(s).

ii. Find the value of K in the system to yield a natural frequency of 7.07 rad/s.

iii. Hence, calculate the system's damping ratio as well as its settling time.

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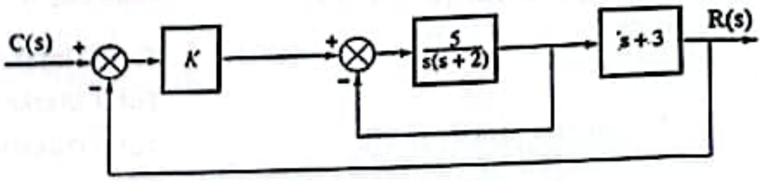
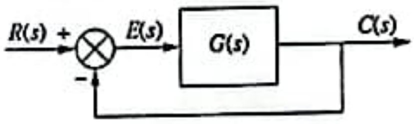


b. For the unity feedback system shown below, where:

$$G(s) = \frac{90}{s^n(s+m)}$$

i. Find the values of n and m to yield a constant steady-state error of 0.01 for a unit-ramp input.

ii. Hence, find the value of n that would eliminate the steady-state error.



CLO # 3: Analyze the stability of a system.

Q3: **[20 marks]**

a. For the unity feedback system shown below, where:

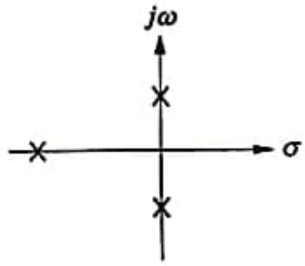
$$G(s) = \frac{K(s+4)}{s(s+1.2)(s+2)}$$

i. Find the closed-loop transfer function, T(s), of the system.

ii. Hence, analyze T(s) to calculate the range of K for stability.

b. Use Routh Hurwitz criteria to calculate K that will place the closed-loop poles of the system as shown in the Figure here; one pole on real-axis and two poles on $j\omega$-axis. The system's closed loop transfer function T(s) is expressed below. Hence, find the value of the poles as well.

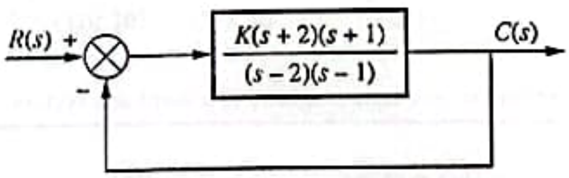
$$T(s) = \frac{2s^4 + (K+2)s^3 + Ks^2}{s^3 + s^2 + 2s + K}$$



CLO # 4: Analyze a given system in frequency domain.

Q4: **[20 marks]**

a) For the system shown in Figure below,



i. Find the break-away and break-in points (if any) of the root-locus.

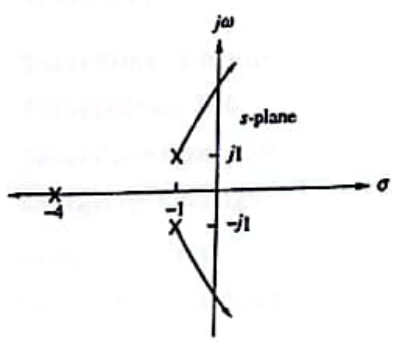
ii. Find the crossings on the $j\omega$-axis of the root-locus.

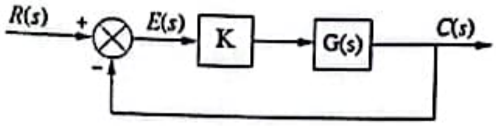
iii. Hence, illustrate the diagram of the root-locus.

b) The root locus of a system, G(s), is shown here:

i. Find the open-loop transfer function, G(s), of the system.

ii. Hence, using the appropriate property, find the gain K for which the system will yield a closed-loop pole on the real-axis at $s = -5$.



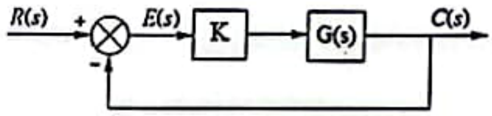


CLO # 5: Design a closed-loop feedback controller

Q5: **[20 marks]**

The open loop transfer function of an un-compensated unity feedback system is given below.

$$K\,G(s) = \frac{K}{(s+2)(s+6)}$$



The dominant pole of the un-compensated system occurs at $K = 29$.

a. Find the location of the dominant pole of the un-compensated system.

b. Find the un-compensated system's settling-time, peak-time, and percentage overshoot.

c. It is desired to compensate the system to decrease its settling-time by a factor of 1.5 without affecting its overshoot. Find the new settling-time, peak-time, and the new dominant pole location to satisfy the required performance metrics.

d. Justify which controller (PI or PD) is best suited for this application.

e. Hence, design the controller selected in (d) to satisfy the desired performance metrics. Write the transfer function of the controller including its pole/zeroes.